

Housing exam 2

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```
library(tidyverse)
library(tsibble)
library(fable)
library(fabletools)
library(feasts)
library(urca)
library(strucchange)
library(scales)
requireNamespace("vars", quietly = TRUE)

price_raw <- read_csv("data/kbh_quarter_sqm_price.csv")
rate_raw <- read_csv("data/bondInt_quarterly.csv")

price <- price_raw |>
  mutate(
    qtr = yearquarter(str_replace(Quarter, "K", " Q")),
    Price = as.numeric(Price)
  ) |>
  select(qtr, Price) |>
  arrange(qtr) |>
  as_tsibble(index = qtr)

rate <- rate_raw |>
  mutate(
    qtr = yearquarter(str_replace(Quarter, "K", " Q")),
    rate = as.numeric(mean_rate)
  ) |>
  select(qtr, rate) |>
```

```

  arrange(qtr) |>
  as_tsibble(index = qtr)

df_raw <- inner_join(price, rate, by = "qtr") |>
  arrange(qtr) |>
  as_tsibble(index = qtr)

cat("Joined sample:", format(min(df_raw$qtr)), "to", format(max(df_raw$qtr)),
    " (n =", nrow(df_raw), ")\n")

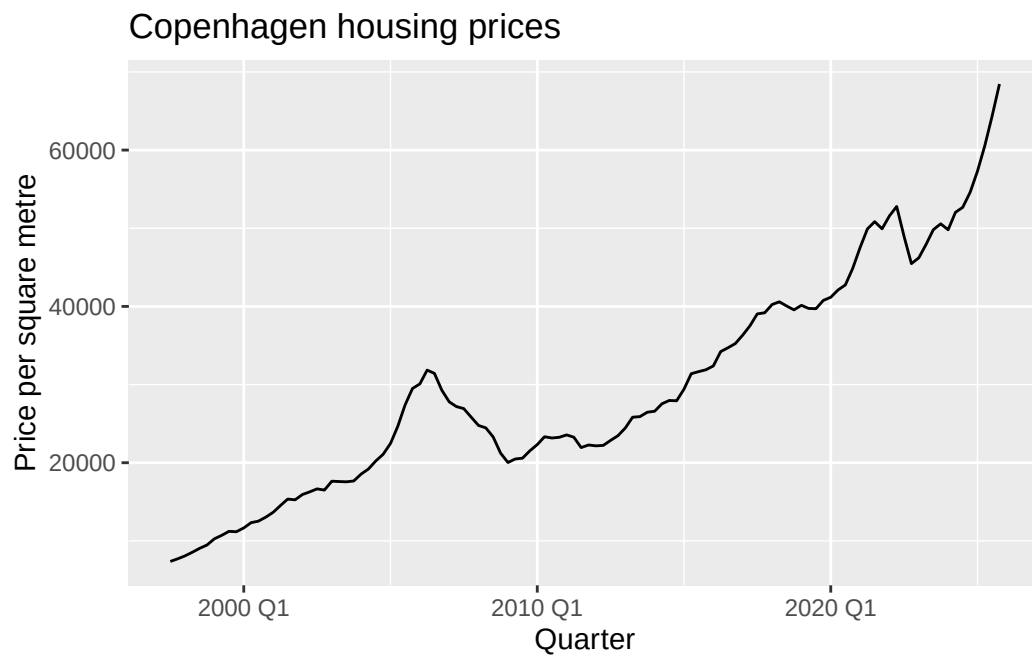
```

Joined sample: 1997 Q3 to 2025 Q4 (n = 114)

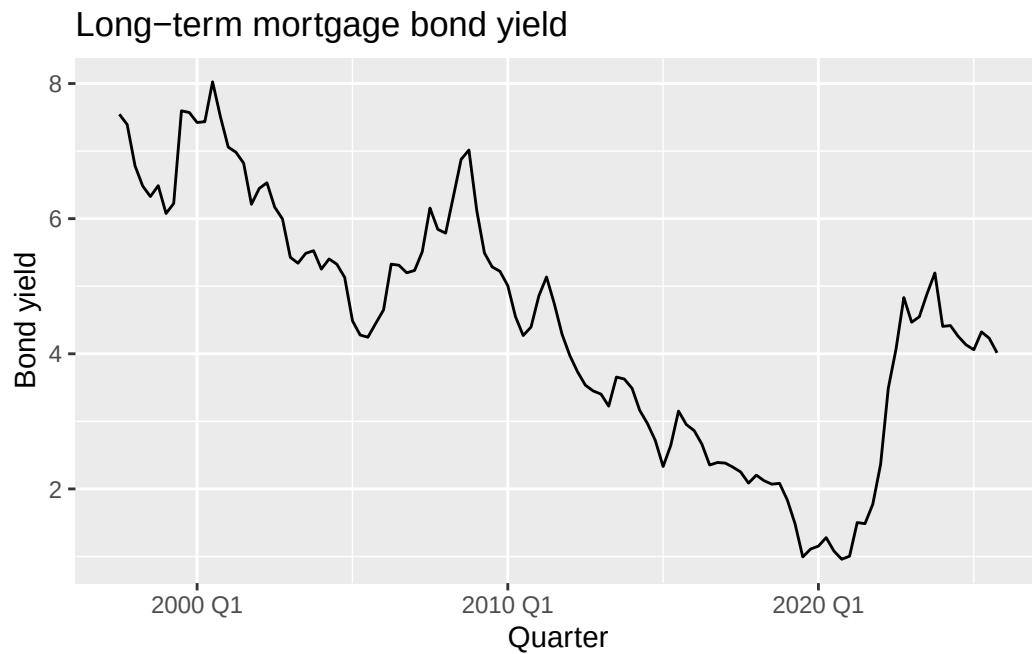
```

df_raw |>
  autoplot(Price) +
  labs(
    title = "Copenhagen housing prices",
    x = "Quarter",
    y = "Price per square metre"
  )

```



```
df_raw |>
  autoplot(rate) +
  labs(
    title = "Long-term mortgage bond yield",
    x = "Quarter",
    y = "Bond yield"
  )
```



```
df_model_raw <- df_raw |>
  mutate(
    rate_lag2 = lag(rate, 2)
  ) |>
  drop_na(rate_lag2)

n_test <- floor(nrow(df_model_raw) * 0.20)

train_raw <- df_model_raw |>
  slice_head(n = nrow(df_model_raw) - n_test)

test_raw <- df_model_raw |>
  slice_tail(n = n_test)
```

```
lambda_train <- train_raw |>
  features(Price, guerrero) |>
  pull(lambda_guerrero)

cat("Box-Cox lambda estimated on training data:", round(lambda_train, 4), "\n")
```

Box-Cox lambda estimated on training data: 0.3916

```
df_model <- df_model_raw |>
  mutate(
    price_bc = box_cox(Price, lambda_train),
    price_bc_diff = difference(price_bc),
    rate_diff = difference(rate)
  )

train <- df_model |>
  slice_head(n = nrow(df_model) - n_test)

test <- df_model |>
  slice_tail(n = n_test)
```

```
train |>
  features(price_bc, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>      <dbl>
1     1.89        0.01
```

```
train |>
  filter(!is.na(price_bc_diff)) |>
  features(price_bc_diff, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>      <dbl>
1     0.179        0.1
```

```
train |>
  features(rate, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>     <dbl>
1     2.04       0.01
```

```
train |>
  filter(!is.na(rate_diff)) |>
  features(rate_diff, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>     <dbl>
1  0.0726       0.1
```

```
train |>
  features(price_bc, unitroot_ndiffs)
```

```
# A tibble: 1 x 1
  ndiffs
    <int>
1      1
```

```
train |>
  features(rate, unitroot_ndiffs)
```

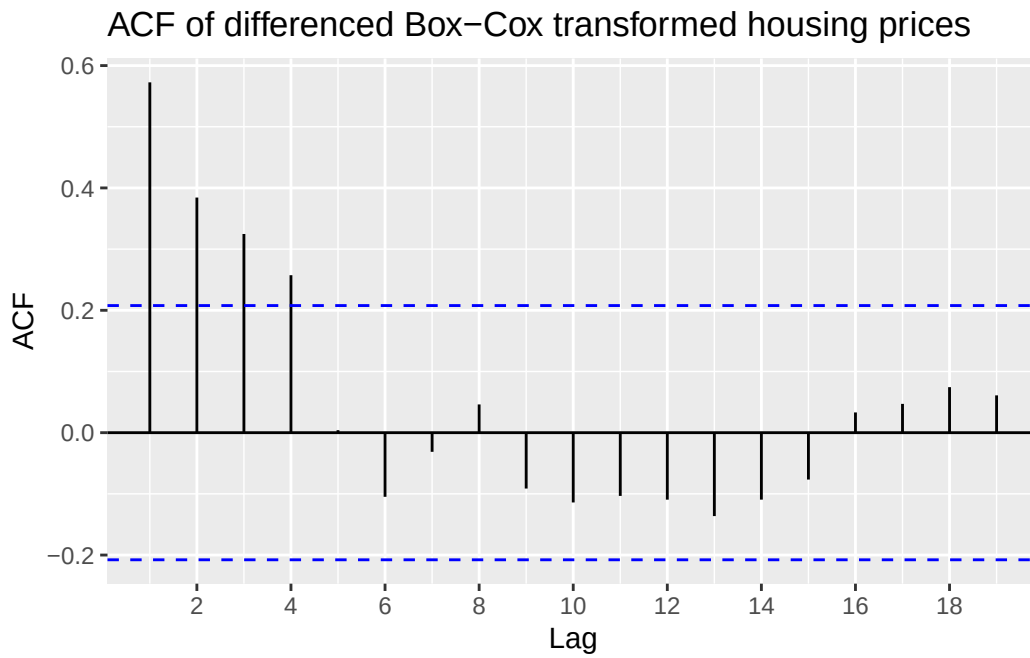
```
# A tibble: 1 x 1
  ndiffs
    <int>
1      1
```

```
train |>
  filter(!is.na(price_bc_diff)) |>
  ACF(price_bc_diff) |>
  autoplot() +
  labs(
```

```

title = "ACF of differenced Box-Cox transformed housing prices",
x = "Lag",
y = "ACF"
)

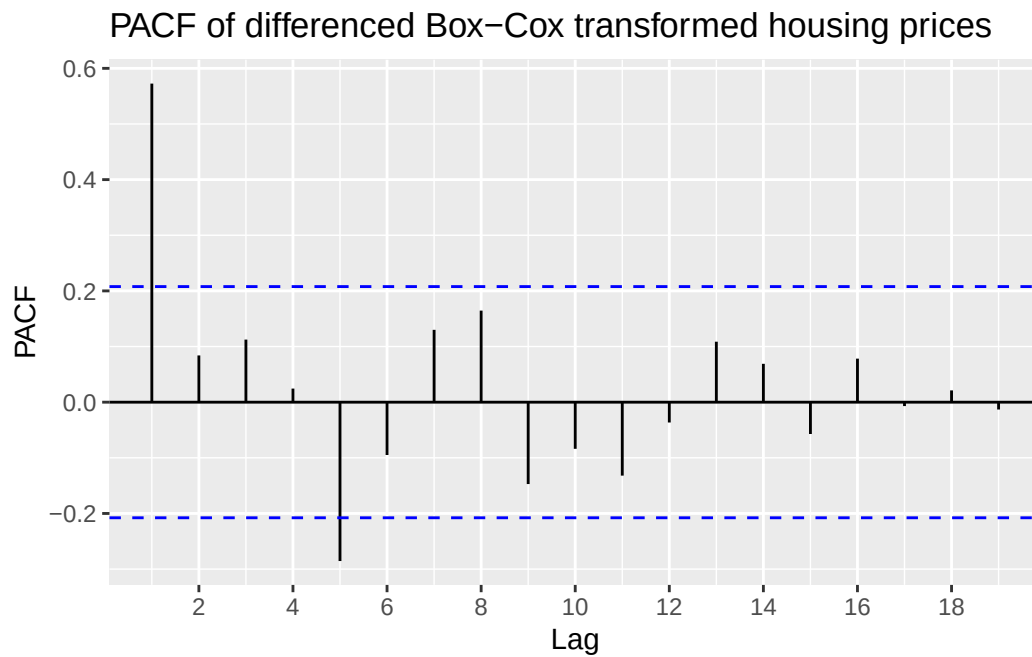
```



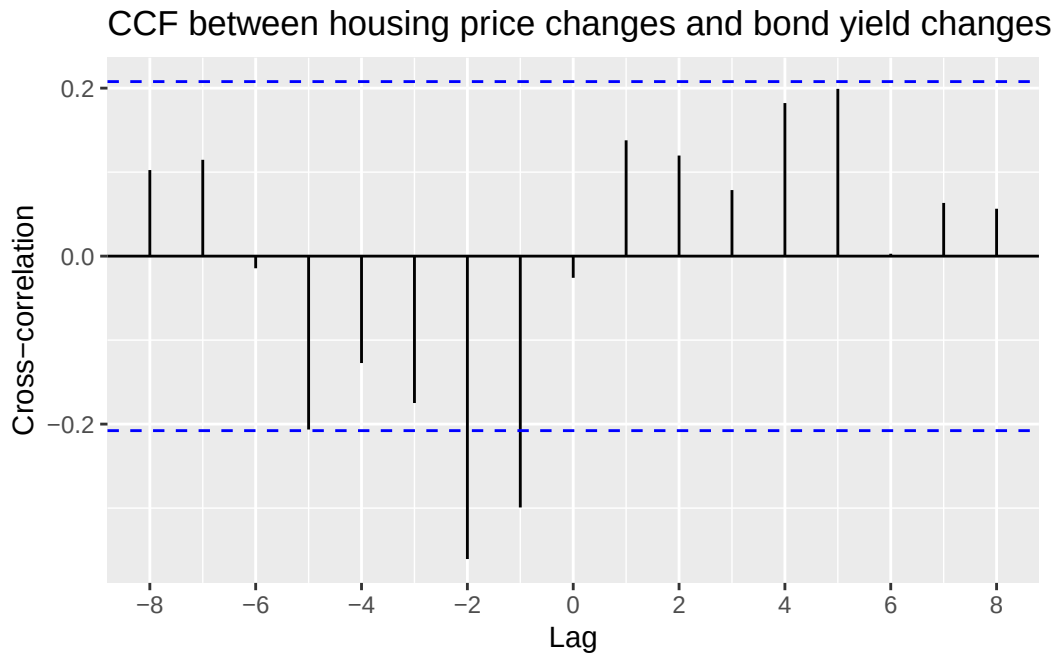
```

train |>
  filter(!is.na(price_bc_diff)) |>
  PACF(price_bc_diff) |>
  autoplot() +
  labs(
    title = "PACF of differenced Box-Cox transformed housing prices",
    x = "Lag",
    y = "PACF"
  )
)

```



```
train |>
  filter(!is.na(price_bc_diff), !is.na(rate_diff)) |>
  CCF(price_bc_diff, rate_diff, lag_max = 8) |>
  autoplot() +
  labs(
    title = "CCF between housing price changes and bond yield changes",
    x = "Lag",
    y = "Cross-correlation"
  )
```



```
fit_models <- train |>
  model(
    arima = ARIMA(price_bc),
    arimax = ARIMA(price_bc ~ rate),
    arimax_lag2 = ARIMA(price_bc ~ rate_lag2)
  )

report(fit_models)
```

```
# A tibble: 3 x 8
  .model      sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
  <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
1 arima        2.14   -159.  326.  326.  336. <cpl [5]> <cpl [0]>
2 arimax        2.20   -164.  342.  343.  359. <cpl [2]> <cpl [8]>
3 arimax_lag2   2.06   -161.  335.  336.  350. <cpl [6]> <cpl [0]>
```

```
fc_models <- fit_models |>
  forecast(new_data = test)

fc_models
```

```
# A fable: 66 x 9 [1Q]
```

```

# Key:      .model [3]
      .model      qtr
      <chr>      <qtr>
1 arima  2020 Q3
2 arima  2020 Q4
3 arima  2021 Q1
4 arima  2021 Q2
5 arima  2021 Q3
6 arima  2021 Q4
7 arima  2022 Q1
8 arima  2022 Q2
9 arima  2022 Q3
10 arima 2022 Q4
# i 56 more rows
# i 7 more variables: price_bc <dbl>, .mean <dbl>, Price <dbl>, rate <dbl>,
#   rate_lag2 <dbl>, price_bc_diff <dbl>, rate_diff <dbl>

```

```

accuracy_models_clean <- accuracy(fc_models, test) |>
  select(.model, RMSE, MAE, MAPE, ACF1) |>
  arrange(RMSE)

accuracy_models_clean

```

```

# A tibble: 3 x 5
  .model      RMSE  MAE  MAPE  ACF1
  <chr>      <dbl> <dbl> <dbl> <dbl>
1 arima        5.21  4.29  2.40 0.735
2 arimax       17.1  14.0  7.72 0.751
3 arimax_lag2  19.1  16.0  8.87 0.764

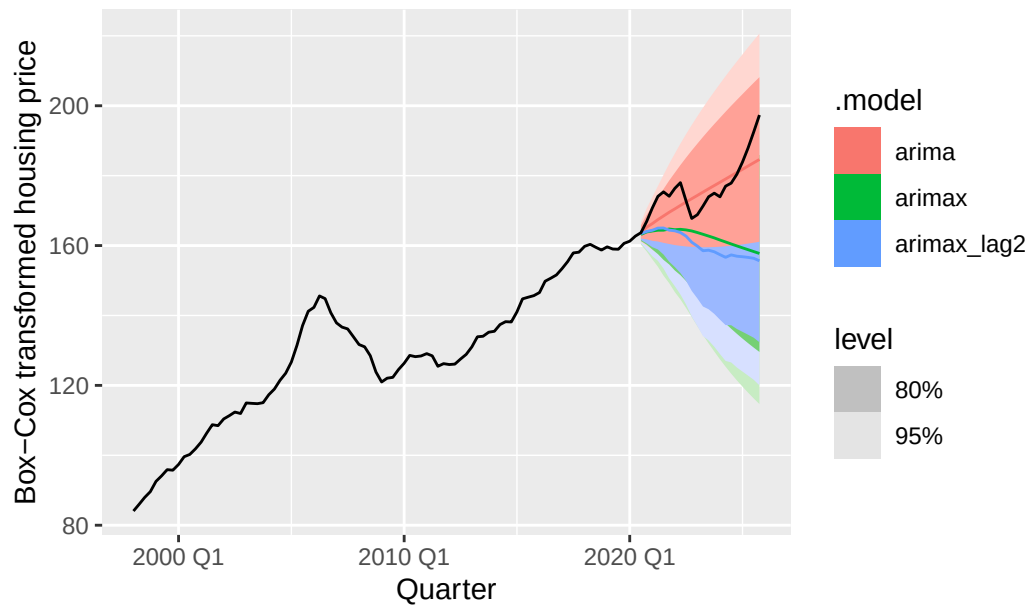
```

```

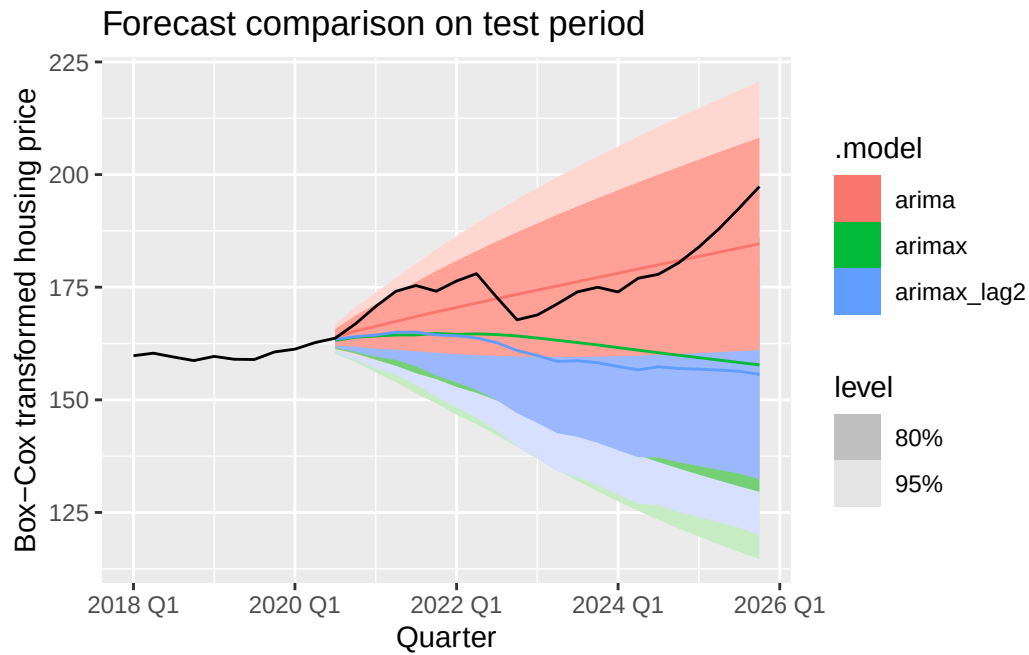
fc_models |>
  autoplot(df_model) +
  labs(
    title = "Forecast comparison: ARIMA vs ARIMAX models",
    x = "Quarter",
    y = "Box-Cox transformed housing price"
  )

```

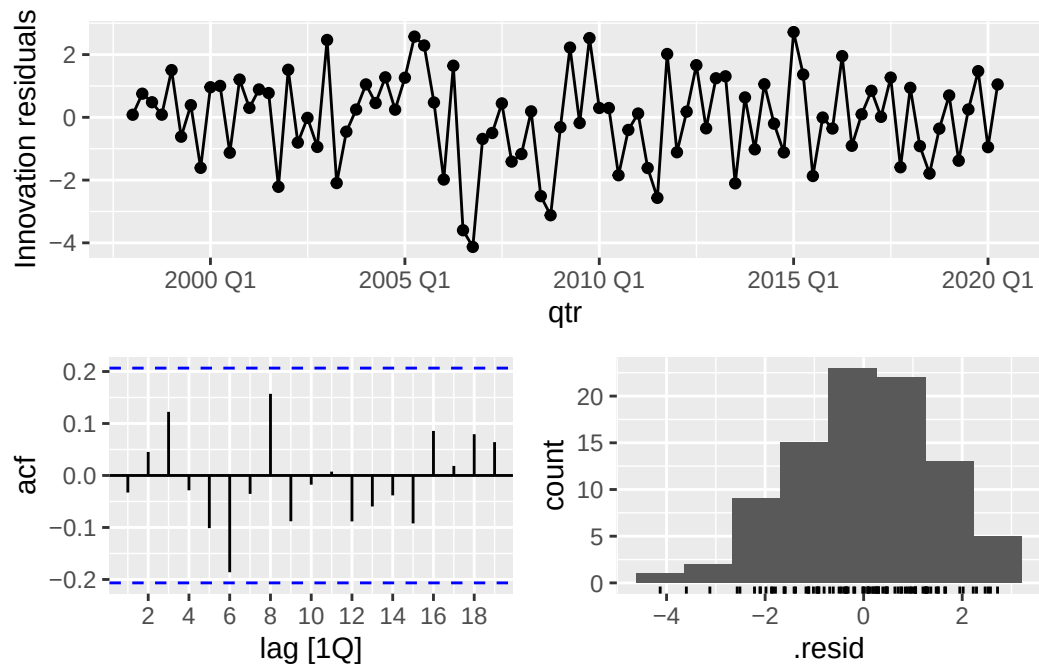
Forecast comparison: ARIMA vs ARIMAX models



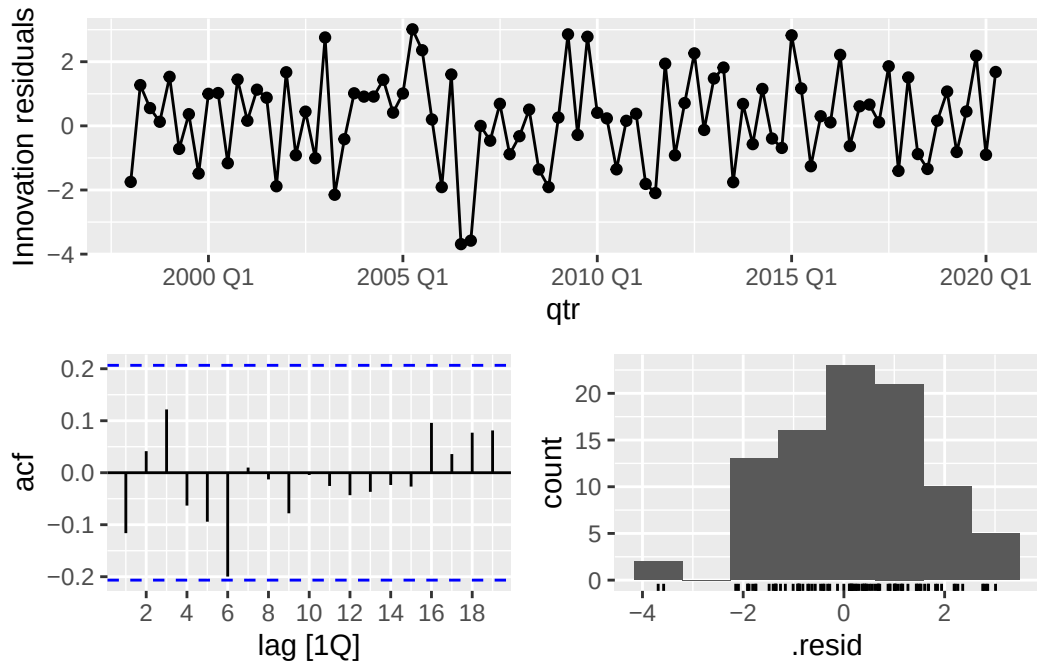
```
fc_models |>
  autoplot(df_model |> filter(qtr >= yearquarter("2018 Q1"))) +
  labs(
    title = "Forecast comparison on test period",
    x = "Quarter",
    y = "Box-Cox transformed housing price"
  )
```



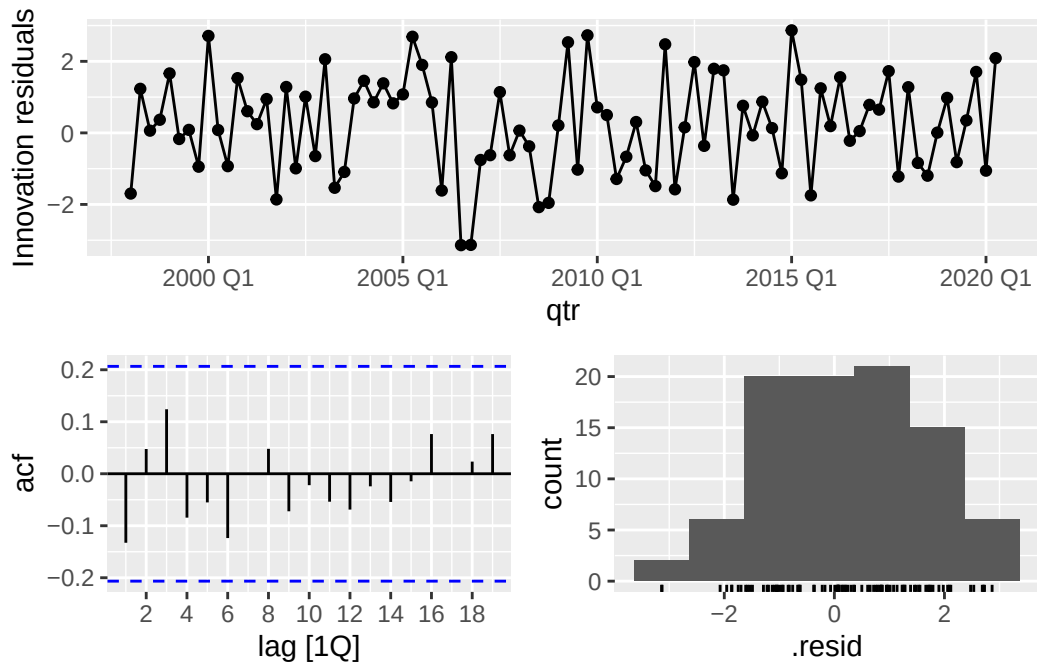
```
fit_models |>
  select(arima) |>
  gg_tsresiduals()
```



```
fit_models |>
  select(arimax) |>
  gg_tsresiduals()
```



```
fit_models |>
  select(arimax_lag2) |>
  gg_tsresiduals()
```



```
fit_models |>
  augment() |>
  features(.innov, ljung_box, lag = 8, dof = 1)
```

```
# A tibble: 3 x 3
  .model    lb_stat lb_pvalue
  <chr>      <dbl>   <dbl>
1 arima      8.83    0.265
2 arimax     8.04    0.329
3 arimax_lag2 6.04    0.535
```

```
var_data <- train |>
  filter(!is.na(price_bc_diff), !is.na(rate_diff)) |>
  as_tibble() |>
  select(price_bc_diff, rate_diff)
```

```
lag_select <- vars::VARselect(var_data, lag.max = 8, type = "const")
lag_select$selection
```

```
AIC(n)  HQ(n)  SC(n) FPE(n)
      2      2      1      2
```

```
p_opt <- lag_select$selection["SC(n)"]

fit_var <- vars::VAR(var_data, p = p_opt, type = "const")
summary(fit_var)
```

VAR Estimation Results:

=====

Endogenous variables: price_bc_diff, rate_diff

Deterministic variables: const

Sample size: 88

Log Likelihood: -180.148

Roots of the characteristic polynomial:

0.4156 0.4156

Call:

vars::VAR(y = var_data, p = p_opt, type = "const")

Estimation results for equation price_bc_diff:

=====

price_bc_diff = price_bc_diff.l1 + rate_diff.l1 + const

	Estimate	Std. Error	t value	Pr(> t)
price_bc_diff.l1	0.56525	0.08323	6.791	1.42e-09 ***
rate_diff.l1	-1.52170	0.44558	-3.415	0.00098 ***
const	0.27901	0.16982	1.643	0.10408

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.417 on 85 degrees of freedom

Multiple R-Squared: 0.4102, Adjusted R-squared: 0.3964

F-statistic: 29.56 on 2 and 85 DF, p-value: 1.795e-10

Estimation results for equation rate_diff:

=====

rate_diff = price_bc_diff.l1 + rate_diff.l1 + const

	Estimate	Std. Error	t value	Pr(> t)
price_bc_diff.l1	0.02703	0.01949	1.387	0.1691
rate_diff.l1	0.23286	0.10434	2.232	0.0283 *

```

const          -0.06796    0.03976  -1.709    0.0911 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 0.3318 on 85 degrees of freedom
Multiple R-Squared: 0.07344, Adjusted R-squared: 0.05164
F-statistic: 3.369 on 2 and 85 DF, p-value: 0.03909

Covariance matrix of residuals:

	price_bc_diff	rate_diff
price_bc_diff	2.00734	-0.02199
rate_diff	-0.02199	0.11006

Correlation matrix of residuals:

	price_bc_diff	rate_diff
price_bc_diff	1.00000	-0.04678
rate_diff	-0.04678	1.00000

```
vars::serial.test(fit_var, lags.pt = 8, type = "PT.asymptotic")
```

Portmanteau Test (asymptotic)

data: Residuals of VAR object fit_var
Chi-squared = 29.833, df = 28, p-value = 0.3713

```

var_forecast <- predict(fit_var, n.ahead = nrow(test))
price_diff_fc <- var_forecast$fcst$price_bc_diff[, "fcst"]

```

```

last_train_price <- tail(train$price_bc, 1)
price_bc_fc <- cumsum(c(last_train_price, price_diff_fc))[-1]

```

```

var_accuracy <- tibble(
  .model = "var",
  RMSE = sqrt(mean((price_bc_fc - test$price_bc)^2)),
  MAE = mean(abs(price_bc_fc - test$price_bc)),
  MAPE = mean(abs((price_bc_fc - test$price_bc) / test$price_bc)) * 100
)

```

```
)
```

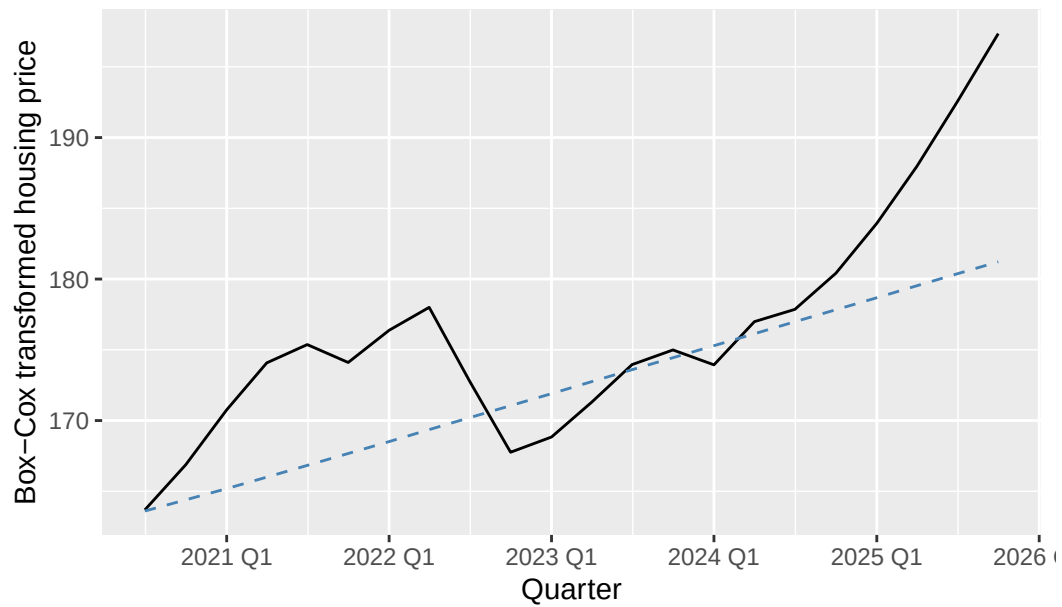
```
bind_rows(accuracy_models_clean, var_accuracy) |>  
  arrange(RMSE)
```

```
# A tibble: 4 x 5
```

	.model	RMSE	MAE	MAPE	ACF1
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	arima	5.21	4.29	2.40	0.735
2	var	6.39	4.85	2.68	NA
3	arimax	17.1	14.0	7.72	0.751
4	arimax_lag2	19.1	16.0	8.87	0.764

```
tibble(  
  qtr = test$qtr,  
  actual = test$price_bc,  
  var_fc = price_bc_fc  
) |>  
  ggplot(aes(x = qtr)) +  
  geom_line(aes(y = actual), colour = "black") +  
  geom_line(aes(y = var_fc), colour = "steelblue", linetype = "dashed") +  
  labs(  
    title = "VAR forecast vs actual (test period)",  
    x = "Quarter",  
    y = "Box-Cox transformed housing price"  
  )
```

VAR forecast vs actual (test period)

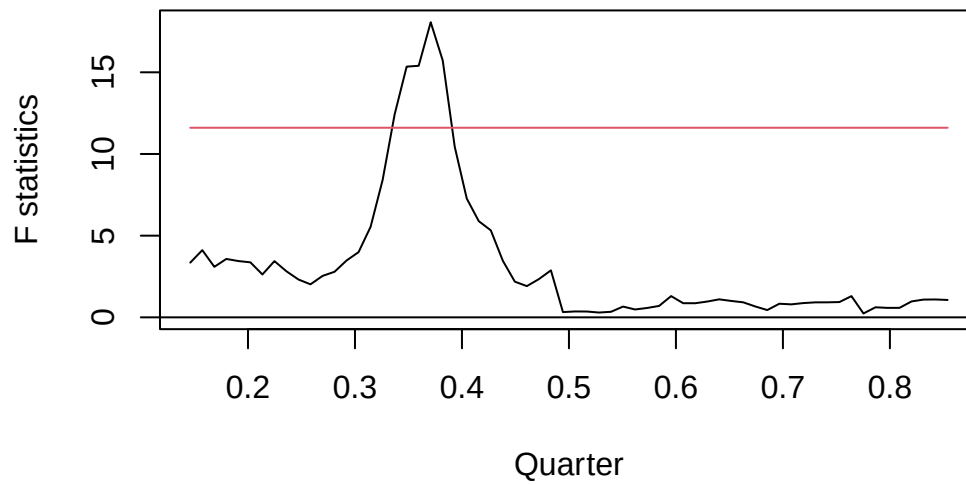


```
qlr_df <- train |>
  filter(!is.na(price_bc_diff), !is.na(rate_diff)) |>
  as_tibble()

qlr <- Fstats(price_bc_diff ~ rate_diff, data = qlr_df)

plot(qlr,
      main = "QLR Test: F-statistics across candidate breakpoints",
      xlab = "Quarter")
```

QLR Test: F-statistics across candidate breakpoints



```
breakpoints(qlr)
```

Optimal 2-segment partition:

Call:

```
breakpoints.Fstats(obj = qlr)
```

Breakpoints at observation number:

33

Corresponding to breakdates:

0.3595506

```
sctest(qlr)
```

supF test

data: qlr

sup.F = 18.065, p-value = 0.002939

```
bp <- breakpoints(qlr)
print(bp)
```

Optimal 2-segment partition:

Call:

```
breakpoints.Fstats(obj = qlr)
```

Breakpoints at observation number:

33

Corresponding to breakdates:

0.3595506

```
qlr_df[bp$breakpoints, "qtr"]
```

A tibble: 1 x 1

qtr

<qtr>

1 2006 Q2

```
break_qtr <- qlr_df[bp$breakpoints, "qtr"] |> pull(qtr)
```

```
train_pre <- train |> filter(qtr <= break_qtr)
```

```
train_post <- train |> filter(qtr > break_qtr)
```

```
fit_pre <- train_pre |>
```

```
  model(arimax_lag2 = ARIMA(price_bc ~ lag(rate, 2)))
```

```
report(fit_pre)
```

Series: price_bc

Model: LM w/ ARIMA(2,0,0)(0,0,1)[4] errors

Coefficients:

	ar1	ar2	sma1	lag(rate, 2)	intercept
	1.6382	-0.6456	0.7708	-1.1503	120.2335
s.e.	0.1339	0.1358	0.2882	0.4888	25.9007

sigma^2 estimated as 1.647: log likelihood=-57.15
AIC=126.31 AICc=129.42 BIC=135.47

```
fit_pre |>  
  augment() |>  
  features(.innov, ljung_box, lag = 8, dof = 1)
```

```
# A tibble: 1 x 3  
  .model      lb_stat lb_pvalue  
  <chr>      <dbl>    <dbl>  
1 arimax_lag2  4.72      0.693
```